



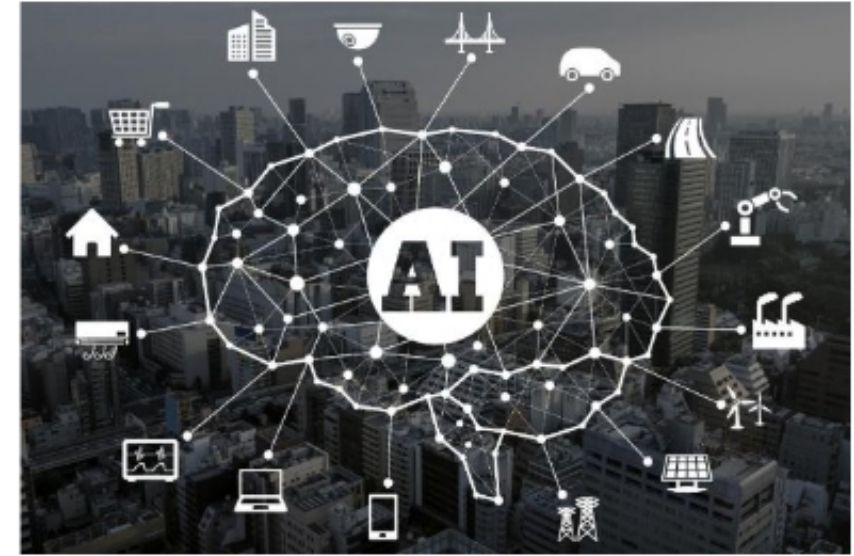
Sujit Gujar

# Artificial Intelligence (AI)

- Artificial Intelligence (AI) is the science and engineering of making machines (including computers, robots and others) intelligent and enabling them to take actions and decisions that humans typically do
- *“Artificial intelligence is one of the most profound undertakings in science, and one that will affect every aspect of human life.”*
  - Stanford President John Hennessy

# Proliferation of AI

- Autonomous Driving, Robotics, Llama
- Computer Vision
- In general AI assisted decision making
  - Crowdsourcing, Crowdfunding
  - Online Advertisements
  - Classification, Jurisdiction, Recruitments
  - Resource Allocation, Course Allocation, Ride Shares, Rent Shares



Ubiquitous AI systems

Image Credits: Manisha Padala Thesis

# Two dimensions to AI

- Human vs Rational
  - Rationality: doing always right thing
- Act vs Think

# Acting Humanly Turing Test

- An human interrogator is not able to distinguish between computer and human after posing some written questions
- Computer need following capabilities
  - Natural language processing
  - Knowledge representation
  - Automated reasoning
  - Machine Learning

Total Turing Test

Involves interaction with objects,  
people,  
Moving objects around etc.,

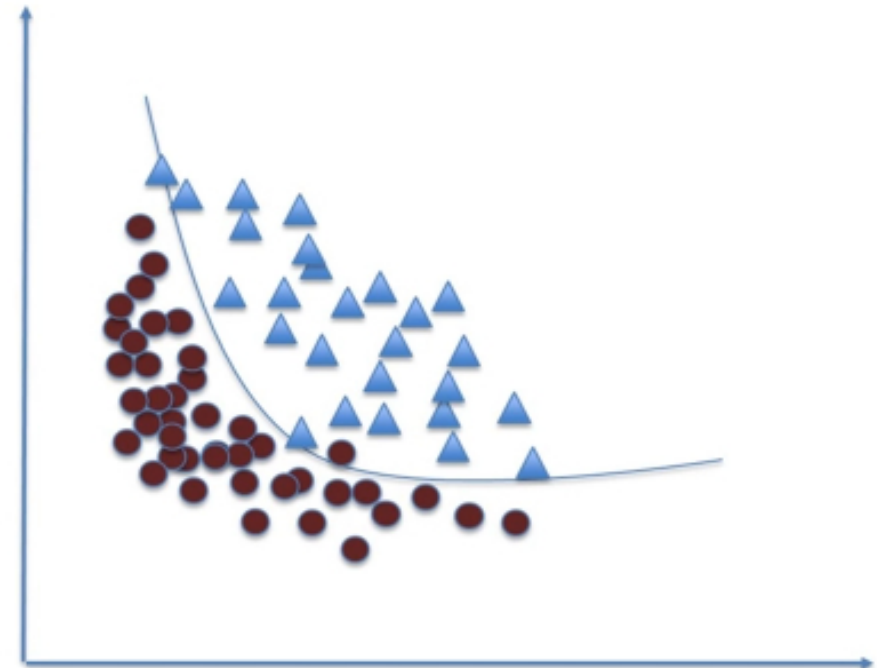
Additionally need Computer Vision and  
Robotics



- Think Humanly
  - Cognitive modelling
- Act Rationally
  - Agent based modelling
  - Agent should operate autonomously, perceive the environment, adapt to change
- Think Rationally
  - Right thinking
  - Logic-based approach
  - Need to involve probability to model uncertain things

# What is Machine Learning?

- Machine learning: construction of algorithms that can **learn** from and make predictions on data
- data == features
- Supervised: Classification, regression
- Unsupervised: Clustering, anomaly detection



# Cricket ball vs tennis ball

Y-Values

180  
160  
140  
120



2.1

2.3

2.5

2.7

2.9

Perceptron: Threshold  
Activation

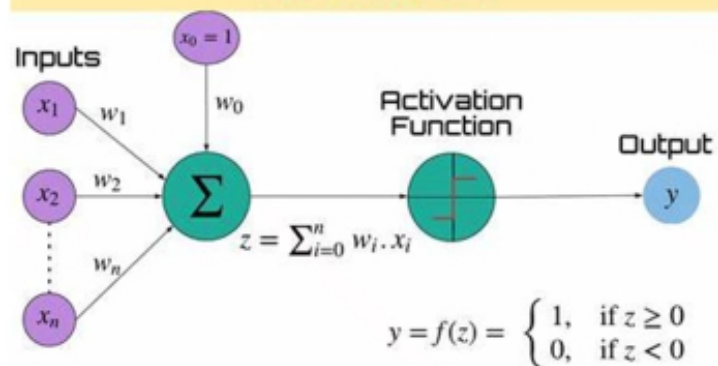
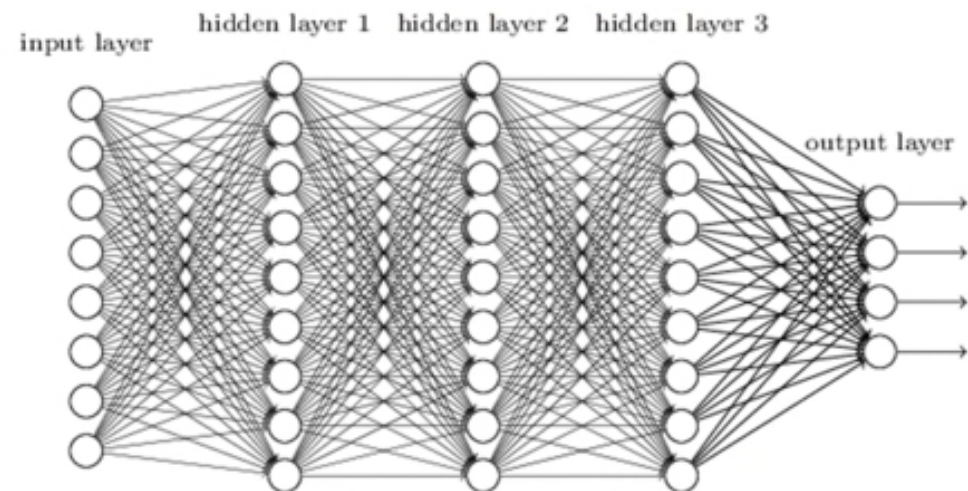


Image:



# Neural Networks and Deep Learning

+Replicate neurons in human brains (Neural Network)





# Complexity of neural networks

GPT 4 (March'23)  
1.76 trillion params

GPT3 May 2021  
175 billion parameters

MT-NLG 530B  
PaLM 540B

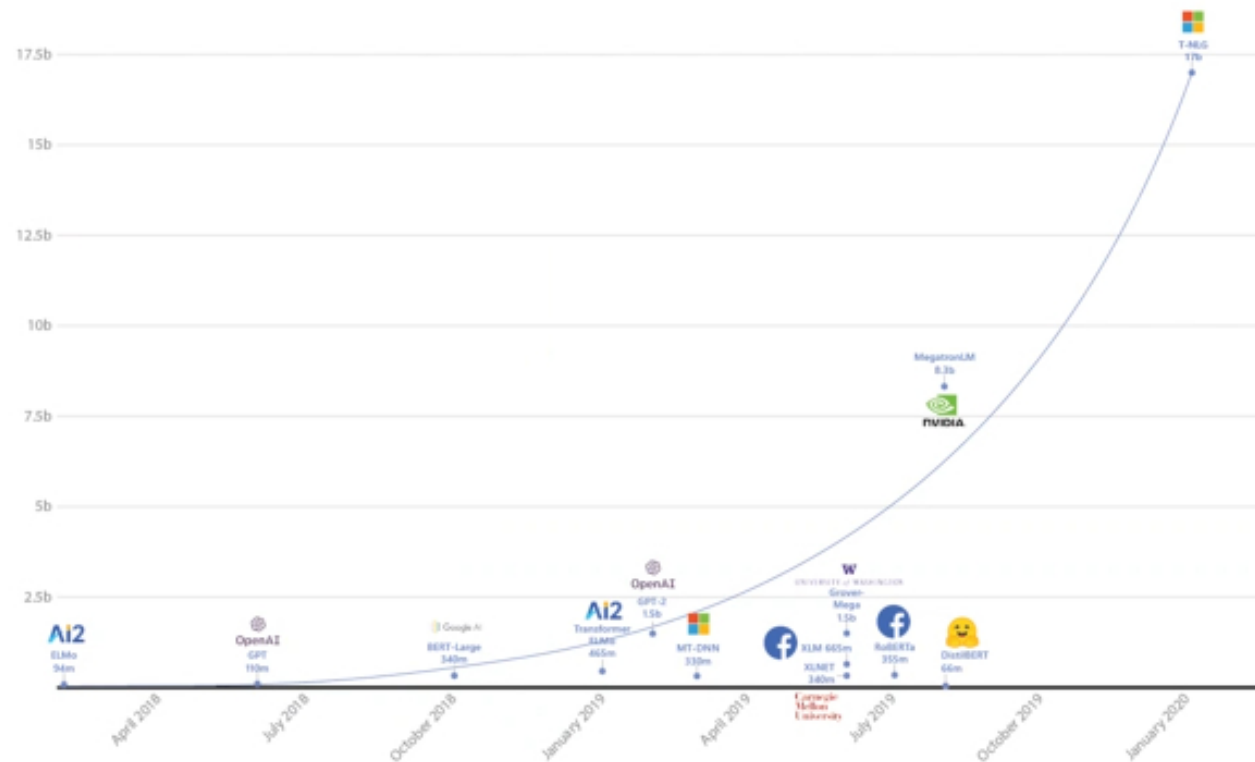


Image credits: [Turing-NLG: A 17-billion-parameter language model by Microsoft - Microsoft Research](#)



# Intelligence

- Intelligence
  - the ability to understand, learn and think
- Intelligence of humans is achieved—not by purely reflex mechanisms but by processes of reasoning that operate on internal representations of knowledge
- Artificial Intelligence (AI) is about building machines and systems that can reason, learn, and solve problems, similar to how humans do. In many AI applications, logical reasoning plays a crucial role, allowing systems to make decisions based on given conditions and facts
- Leads to knowledge-based agents approach for AI

# Knowledge Base

- Knowledge Base – Set of sentences represented in knowledge representation language
- Knowledge representation language--  
Expressing knowledge explicitly in a computer-tractable way
  - (logic)

## Logic

- Oxford dictionary
  - a way of thinking or explaining something
  - sensible reasons for doing something
  - the science of thinking about or explaining the reason for something using formal methods
- Logic is the basis of all mathematical reasoning and all automated reasoning

